

Investigating Spline and Triangular Reinforced Flexures

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Abstract—This study analyses the effect of triangular and spline reinforcements on leaf springs. It focuses on maximizing the in-plane displacement referred to as the Range of Motion (RoM), while also maximizing the out-of-plane stiffness. This was achieved using the finite element simulations in Solidworks and MATLAB. A reference model has also been simulated for different triangle heights for comparison. Finally, the 3D-printed models were experimentally tested to fully understand the design. From this, it can be concluded that the RoM can be increased by 6% if some of the triangular structures are replaced with bows. This however, will also negatively impact the stiffness resulting in a decrease of 22%. A trade-off between RoM and buckling may be acceptable if it fits the configuration better.

I. INTRODUCTION

Compliant mechanisms (CMs) enable movement without friction and backlash [1]. Compared to traditional rigid-body mechanisms, which gain movement through rolling or sliding contacts between two parts, CMs are ideal for small and precise movements. They gain mobility through elastic deformations and are monolithic [1]. A downside of CMs is the limited range of motion (RoM). The movement is blocked by the structure itself, because CMs lack joints. Additionally, the RoM is limited by plastic deformation which is determined by the material properties. Plastic deformation results in non-repeatable movement. This study focuses on the effect of reinforcement structures on desired and undesired stiffness of CMs that use leaf springs as their deflective members.

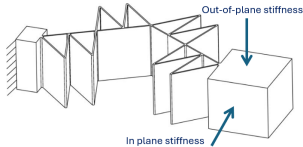


Fig. 1: Leaf spring with triangular reinforcement structures. Figure taken from [2]

An issue of a leaf spring is that it becomes inherently susceptible to geometric non-linearities and buckling behaviour, becoming more vulnerable to torsion and out-of-plane deflection (buckling). A good way to combat these undesired effects is to add torsion reinforcement structures to the leaf springs ([2][3][4] and Figure 1). These structures prevent torsional movement parallel to the leaf spring axes and increase out-of-plane stiffness. However, as a side effect, they also increase stiffness in the desired in-plane movement (RoM), making the leaf springs less flexible. The structures are also prone to colliding with themselves. These factors can make it more

difficult to bend the leaf springs and could decrease the potential RoM of the CM.

This design study has been conducted as part of the Bachelor End Project for Mechanical Engineering students at the TU Delft in the third year of their Bachelor studies.

II. MOTIVATION

The reference paper [2] implements triangular reinforcement structures as a solution to combat out-of-plane deflection and torsion. The design choice of a triangular shape in the reference is not supported [3]. A higher out-of-plane stiffness results in a stiffer mechanism as a whole, which can also result in a higher in-plane stiffness. The reference is optimized for high out-of-plane stiffness. It is essential for many appliances to have high out-of-plane stiffness, but there are also many appliances where a high RoM is required. An example for a high RoM mechanism is a wafer handler. Currently, only the effects of triangular reinforcements are studied. Other shapes like splines have not been considered yet.

Triangles increase the torsional and out-of-plane stiffness [5], but also limit the desired in-plane bending stiffness. Other reinforcements, such as splines, are not considered. These could potentially allow more flexibility in the mechanism. It would be valuable to analyse the impact of triangles and splines on the in- and out-of-plane stiffness.

To investigate the effect, triangular reinforcements are replaced with bent counterparts. The effects of these design changes are compared to the triangular reference using FEM and experiments.

III. DESIGN

The implementation of a bow should give more RoM, but also more out-of-plane buckling, based on intuitive reasoning. The reason for this is the curvature of the bow w.r.t. a triangle. A bow is smoother and continuous. This can result in a better distribution of the stress, which makes the structure more flexible. A triangle has a sharp edge that creates two short flexures that offer more bending stiffness. There is a trade-off between RoM and buckling. If a mechanism is stiffer, this results in less buckling and also less RoM, and vice versa for a flexible mechanism. This trade-off suggests that the design can be tuned to application specific properties. There could also be an optimum ratio if the trade-offs scale non-uniformly.

To increase the RoM the design in Figure 2 is proposed. This geometry will be optimized for a maximized RoM and minimized buckling and the results will be compared to the reference [2].

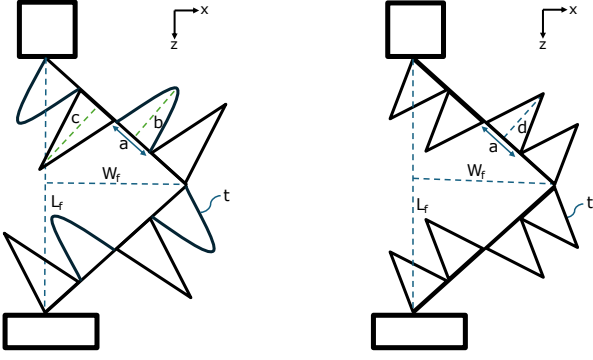


Fig. 2: Proposed design (left) and reference design (right) with variables (b, c and d)

This study investigates the impact of the height (b , c in Figure 2) of a shape has on the buckling and RoM, and how a change of height affects the structure's properties. The hypothesis is that increasing height (adding material) will result in more stiffness, because the extra material added will resist more against deformation.

The expectation of the proposed design is that the combination of bows and triangles will lead to much more RoM with a little more buckling.

IV. SIMULATION INPUT

Optimizations and analyses have been performed in two different programs. One set of simulations has been done in SolidWorks and another has been attempted in MATLAB.

In SolidWorks, the *Design Study* function was used to iteratively optimize the deflection in x and y directions by varying the height of the bows and triangles. Simulations were performed for the different reinforcement structures with a baseline measurement of the reference model [2].

The raw data can be found in the Appendix.

Simulations with a full bow design resulted in excessive stress concentrations within reasonable dimensions. A bow height of 69mm has to be used to gain stress concentrations within the limit. This value is out of range for the parameters as to avoid collision during deflection, so this model will not be considered. This is because a spline with a large height approaches a triangular shape. The data is available in Simulation 3 from the Appendix.

A simplified analytical model has also been constructed in MATLAB. Making use of the *Global Optimization Toolbox* in order to create a faster optimisation method than the *Design Study* function in SolidWorks. However, due to the inherent complications caused by the shape of the flexure, the deflection cannot be approached with an analytical equation. Hence the analytical model does not closely reflect the FEM simulations performed in SolidWorks as can be seen in Figure 3.

Figure 3, also contains the results of a simple cantilever beam SolidWorks model, having the same amount of material, but with a simplified shape. It can be seen that the MATLAB

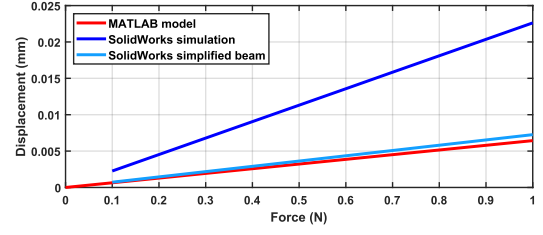


Fig. 3: MATLAB results compared to SolidWorks FEM

model more closely resembles this simplified model. Seeing as a lot of the material is not inline with the fixture and end-effector point in our reinforced flexure like it is with the cantilever beam, we can assume that this is root cause of the mismatch between FEM and MATLAB.

A. Constraints

To ensure a high life cycle and no contamination from particles, the maximum stress of the optimized design should be below 50% of the yield strength to make sure ample leeway exists before the structure can plastically deform. The triangle height (c) and the bow height (b) are parameters to be determined. The maximum height of the bow and triangle are 50 mm and 60 mm respectively. The maximum height of a bow is determined to be 50 mm, since larger values approach triangular geometries and SolidWorks has trouble extruding these features. The maximum heights are chosen to ensure a similar footprint to the reference and no collision during deformation. All the other variables, such as flexure length and -width, are kept the same to the reference to ensure a good comparison.

TABLE I: Constraints optimization

Parameter	σ_{max}	L_f [mm]	a [mm]	b [mm]	c [mm]	d [mm]
Bound	$< 0.5 \cdot \sigma_{yield}$	150	10.94	15-60	15-50	41

B. Parameters

Since the prototypes are built from PETG using FDM printing, the simulations are also performed with PETG to allow for better comparison.

TABLE II: Parameters PETG

Parameter	σ_{yield}	force (x)	force (y)	height flexure	thickness
Value	53 MPa	0.5 N	-0.5 N	50 mm	1 mm

The thickness of the flexures will be set to 1 mm due to the nozzle size limitations of the FDM printer. To accommodate for the thickness and material changes compared to the reference model, the force applied is lowered.

V. SIMULATION OUTPUT

A. Results of simulations

To evaluate the performance of our designs, we performed a range of simulations where both the triangle height and bow height were varied. The goal was to determine the

relation between the range and buckling of the system and its configuration.

Figure 4 illustrates the score for the RoM and buckling as a function of the triangle height, for several different bow heights (b -values). A higher score indicates a higher RoM or a lower buckling w.r.t. the triangular reference. This reference point is marked by the large red $\times$ symbol on the plot and serves as the reference for score calculation. Figure 4 is the data from simulation 1 and 2 from the Appendix.

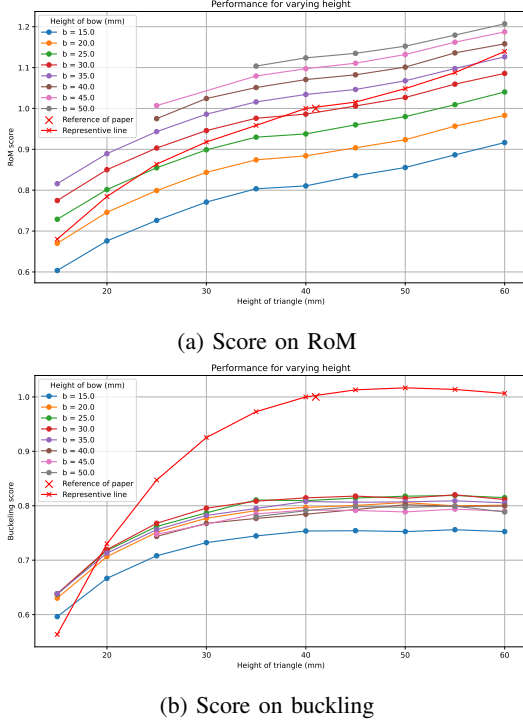


Fig. 4: Performance results

The red line marked with crosses is constructed by varying the height of the triangles of the reference, marked by a single cross point in Figure 4, in our simulation. These data points serve as a reference to highlight the added value of implementing bow structures.

B. Observations and recommendations

From the results, we observe a consistent trend: Increasing the height of the triangles and/or bows increases the RoM. Depending on the requirements, this decision could fit better than making it solely out of triangles.

The performance of the buckling seems to stagnate after a height of 40 mm for the triangles in both the only triangle and the bow and triangle configuration while around this point is also where the transition point of the RoM is. This is likely, because the support structures become so long, they start to become significantly less stiff. This then also allows the leaf spring to bend more easily and have a torsional problem of their own. This result matches with the value of the reference paper[2]. The reference is optimized for minimum buckling,

which is why a value of 41 mm is chosen. For RoM however, it is advisable to increase the height of the triangle.

The bow triangle variant scores around 6 percent better than the triangle only variant in RoM, but performs around 22 percent worse in the buckling. The impact of implementing bows is definitely visible in this comparison. In conclusion, bows increase RoM, but also increase buckling. The buckling seems to be influenced more than the RoM. So what configuration should one choose?

This depends on the priorities. Triangles are useful against buckling, and the heights increase the RoM. The designer can tune the parameters for their application.

VI. EXPERIMENTAL VALIDATION

Alongside the simulations, experiments are performed to verify these findings. The RoM will be measured to compare with the FEM analysis. However, the buckling will not be measured, as the expected deformation is below one millimetre and the equipment necessary for precise measurement at this scale is not available within the time frame.

Two models have been printed on a similar range score and tested. The first model is based on the reference with a triangle height of 41 mm. This received a score around 1. The second model is with a bow height of 23 and a triangle height of 49 mm and received a range score around 1 (see Figure 5). The second model was the result of an early optimization, for more elaboration on this simulation, we refer to Simulation 4 in the Appendix.

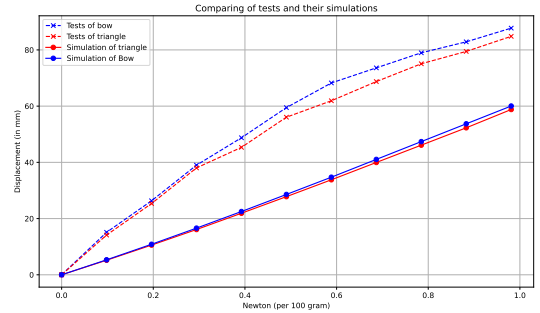


Fig. 5: Comparing two models with similar scores

The experiments show an almost linear relationship. After a force of 0.5N the results start to deviate w.r.t. each other. The simulations show an almost perfect linear relationship. The stress in all simulations for forces until 1N are below 50% of the yield strength, but due to print imperfections, it is a possibility that both mechanisms have undergone plastic deformation. There is a magnitude difference between the simulations and experiments and this could be due to friction. Print imperfections have an impact on the geometry and the non-linear behaviour which can cause a difference as well. In the corners of the supporting structures, fillets have been added to distribute the stress.

The prototypes are secured in the test setup (see Figure 6). A baseline measurement is taken for both to create a starting point. After which, forces are applied from 0N to 1N in steps

of $0.1N$. The force is applied in the x-direction on the end of the carriage.

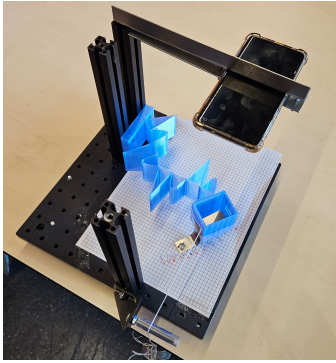


Fig. 6: Test setup

There are some factors of the test setup which can lead to discrepancies in the results. For one, there is a noticeable amount of friction in the roller, which can lead to less RoM. Additionally, the prints are layered and dependent on the printing direction. Thus, the material is not a perfectly solid monolithic structure like the simulation models. The prints will also have slight deviations in their dimensions. These deviations are especially critical for the fillets in the model. Because the space between the support structures is small, imperfections in this area can have a large effect on the displacement. All these factors have an effect on the experimental results, with the fillets deviation having the biggest impact.

VII. DISCUSSION

The combination of the triangular and bow support structures behaves as expected. It utilizes the bow's flexibility at the expense of reduced out-of-plane stiffness. The implementation of bow structures leads to a reduction of in-plane stiffness for triangles bigger than 30mm, while also reducing the out-of-plane stiffness. So it is logical that the proposed design has similar results with the reference. The high buckling in the proposed design can be explained by the absence of the sharp edge from the triangle. This sharp edge seems to be very critical in resisting buckling and it also does not seem to resist the RoM too much in comparison to bows.

In general, the RoM is higher by increasing the height of the shape. The reinforcements are also leaf springs themselves and as they become longer, their stiffness decreases as a result. This allows them to bend more, which allows the main leaf spring to also bend more. The stress can be more evenly distributed along the shapes if there is more material in the shapes. This improves the RoM. But this is not the case for the buckling. Here there seems to be a convergence, after which the buckling does barely change. We cannot justify why there is a convergence in this case.

Even though the test results do not fully accurately represent the SolidWorks simulations due to the aforementioned factors, the relation between the two models seem to be similar for the test compared to the simulations. One factor this study does not consider, is the torsional rotations of the leaf springs.

Torsion can be an important factor for certain applications, so studies would have to be done to research how these changes affect the torsion. Another factor that seemed to have a rather large effect on the deformations of the leaf spring were the fillets. The fillet sizes seemed to quite noticeably change the deformations. It would be interesting to further study the impact of fillets on the RoM and buckling. The buckling is not experimentally validated, because of deformations on a scale of $10\text{ }\mu\text{m}$. FEM is accurate and the FEM analysis on the RoM is confirmed by the experiments, so it is plausible that the FEM analysis on the buckling is reasonable.

VIII. CONCLUSION

In this study, the effects of differently shaped torsion reinforcement structures for leaf springs are presented. A design has been proposed, which incorporates a combination of bows and triangles as the supporting structures to the leaf spring. This combines the high stiffness and strength of triangles, with the low stiffness and flexibility of the bows. The height of these bows and triangles has been iteratively varied within certain boundaries to investigate the effects of the support height. These iterations are compared to a reference point, which consists of a model with only triangles at a height of 41mm. For high bow and triangle heights, the RoM can be increased up to 20% w.r.t. the reference point. This reference model has also been simulated for different triangle heights. Increasing triangle size above 41mm increases the RoM, while not substantially increasing the buckling. To increase the RoM even further without increasing the area, bows can be implemented. An extra RoM of up to 7% can be gained through this. However, this also increases the out-of-plane deflection by up to 22%. Overall bows can be implemented to increase the RoM without increasing the total area, with the trade-off being that it also increases the out-of-plane deflection by approximately 20%.

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